Donut iso variants: inner-R truth-purity scan and ABCD closure

All-calo 120 MeV donut with $R_{\text{outer}} = 0.4$ (cross-check)

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Executive summary

Two new reco-isolation definitions were added as cross-check variants (`syst_type=None`, `syst_role=None`):

- **donutFull** (`use_topo_iso=4`): `cluster_iso_04` − `cluster_iso_{005,0075,02}`. All-calo 120 MeV tower sum (EMCal+HCalIn+HCalOut), outer $R = 0.4$, inner $R \in \{0.05, 0.075, 0.20\}$. Cluster ET is subtracted as a scalar in both terms and cancels in the difference.
- **donutExcl** (`use_topo_iso=5`): `cluster_iso_excl_04` − `cluster_iso_excl_{005,0075,01,02}`. Same three-calo 120 MeV sum, but EMCal towers owned by the cluster are skipped at tower granularity rather than subtracted as a scalar. Inner $R \in \{0.05, 0.075, 0.10, 0.20\}$.

Switch added in `RecoEffCalculator_TTreeReader.C:2042-2065`, mirrored in five companion macros. `donutFull_01` is deliberately skipped because `cluster_iso_01` does not exist in the slimtree (logged in `reports/anatreemaker_reprocess_TODO.md`, along with the EMCal-only inner-ring branches needed for a pure-EMCal donut).

Key finding: `donutFull_02` and `donutExcl_02` give $\sim 2.5\times$ better ABCD-on-MC closure than nominal topo04 iso (mean closure deviation $\langle |1 - r| \rangle \approx 0.07$ vs. nominal 0.185). Low-pT closure is worst across *all* variants (ratio 1.35–1.80 at 9 GeV), high-pT closure is 0.83–0.87 at 34 GeV.

Two-pass workflow

Per-variant iso cut was derived at 80 % target efficiency:

1. **Pass 1**: run the full `RecoEffCalculator_TTreeReader + MergeSim + CalculatePhotonYield` with placeholder loose cut ($b = 100, s = 0$).
2. **Cut derivation**: parameterized `FindETCut.C(var_type, target_eff)` fits $E_T^{\text{iso,cut}} = b + s\,p_T$ at 80 %; `derive_iso_cuts.py` drives all 7 variants.
3. **Pass 2**: regenerate configs with derived (b, s), re-run pipeline on HTCondor cluster 2701654.

Table 1: Derived 80 % iso cuts (isoET $< b + s\,p_T$).

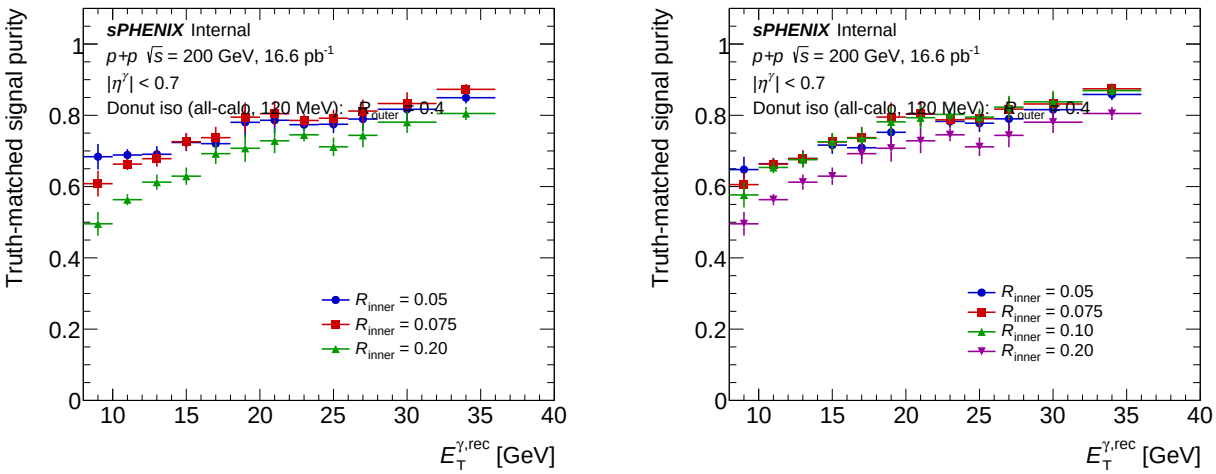
Variant	b [GeV]	s
donutFull_005	0.1089	0.0224
donutFull_0075	0.2642	0.0061
donutFull_02	0.2305	−0.0012
donutExcl_005	0.2457	0.0111
donutExcl_0075	0.2774	0.0053
donutExcl_01	0.2921	0.0017
donutExcl_02	0.2305	−0.0012

Truth-matched signal purity

Purity read from `Photon_final_bdt_<var>_mc.root` (MC-as-data, truth labels populated).

Table 2: Truth-matched photon purity at three representative p_T bins. Nominal topo04 iso included for reference.

Variant	$y(9\,\mathrm{GeV})$	$y(21\,\mathrm{GeV})$	$y(34\,\mathrm{GeV})$
nom (reference)	0.581	0.743	0.797
donutFull_005	0.684	0.786	0.849
donutFull_0075	0.608	0.805	0.873
donutFull_02	0.496	0.728	0.805
donutExcl_005	0.648	0.803	0.858
donutExcl_0075	0.606	0.804	0.874
donutExcl_01	0.577	0.793	0.869
donutExcl_02	0.496	0.728	0.805



(a) donutFull family.

(b) donutExcl family.

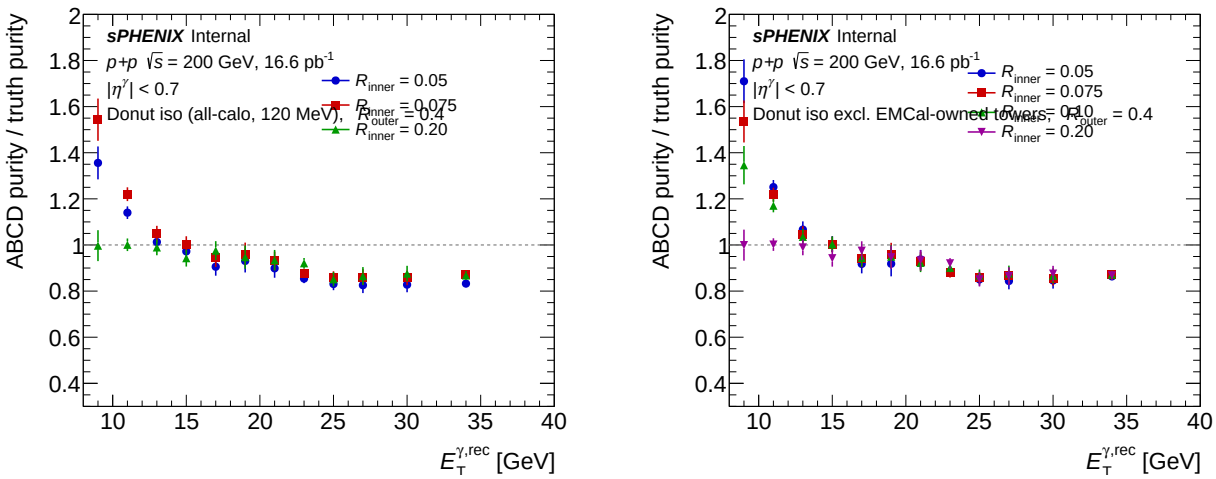
Figure 1: Truth-matched signal purity vs. cluster p_T for the donut-iso variants. Tighter inner R (smaller value) yields higher purity at low p_T; variants converge at high p_T.

ABCD-on-MC closure

The key validation: does the ABCD background subtraction reproduce the truth purity on MC? The ratio $r = \mathrm{purity}_{\mathrm{ABCD}}/\mathrm{purity}_{\mathrm{truth}}$ is stored in `g_mc_purity_fit_ratio`; $r = 1$ is perfect closure.

Table 3: ABCD/truth closure ratio at selected p_T bins and mean absolute deviation from unity. Lower $\langle|1-r|\rangle$ means better closure.

Variant	$r(9)$	$r(15)$	$r(21)$	$r(28)$	$r(34)$	$\langle\|1-r\|\rangle$
nom (reference)	1.795	0.972	0.910	0.842	0.832	0.185
donutFull_005	1.356	0.972	0.899	0.826	0.833	0.136
donutFull_0075	1.543	1.004	0.931	0.861	0.873	0.138
donutFull_02	**0.997**	**0.943**	**0.934**	**0.865**	**0.868**	**0.069**
donutExcl_005	1.710	1.004	0.937	0.844	0.864	0.164
donutExcl_0075	1.536	1.002	0.929	0.866	0.874	0.137
donutExcl_01	1.346	1.003	0.922	0.877	0.872	0.113
donutExcl_02	**0.999**	**0.942**	**0.933**	**0.865**	**0.868**	**0.070**



(a) donutFull family. (b) donutExcl family.

Figure 2: Closure of ABCD-derived purity against truth-matched purity on MC. The dashed line at 1.0 is perfect closure. All variants (including nominal) show a common low-p_T over-prediction and high-p_T under-prediction; inner $R = 0.20$ variants are closest to unity across the full p_T range.

Physics observations

- **Closure is the headline result.** The nominal topo04 iso has an 18.5% average deviation from unity; donut variants with $R_{\text{inner}} = 0.20$ reduce this to 6.9–7.0%. Other donut variants fall in between (11–16%).
- **Low-p_T non-closure is universal.** At $p_T = 9\,\text{GeV}$ the ratio is 1.35–1.80 across *all* variants including nominal. This is probably driven by the fit-region statistics at the edge of the reco p_T binning, not by the iso definition itself.
- **High-p_T under-prediction is also universal**, with $r \approx 0.83$–0.87 at $34\,\text{GeV}$ across all variants. ABCD consistently under-estimates truth purity at high p_T regardless of the iso definition.
- **Inner R dominates purity, not outer-cone treatment.** donutFull vs. donutExcl at matched R_{inner} give truth purities within ± 0.05 at low p_T and $\lesssim 0.02$ at high p_T. `donutFull_02` and `donutExcl_02` are numerically identical to four decimals because the fitted iso cut is the same (see derived (b, s)).

- **Tighter inner R gives higher truth purity at low p_T.** For the Excl family, $y(9\,\mathrm{GeV})$ monotonically decreases from 0.648 at $R_{\mathrm{inner}} = 0.05$ to 0.496 at $R_{\mathrm{inner}} = 0.20$. Interpretation: larger inner R admits less annulus area, giving a looser absolute iso cut that keeps more background.

Caveats and follow-ups

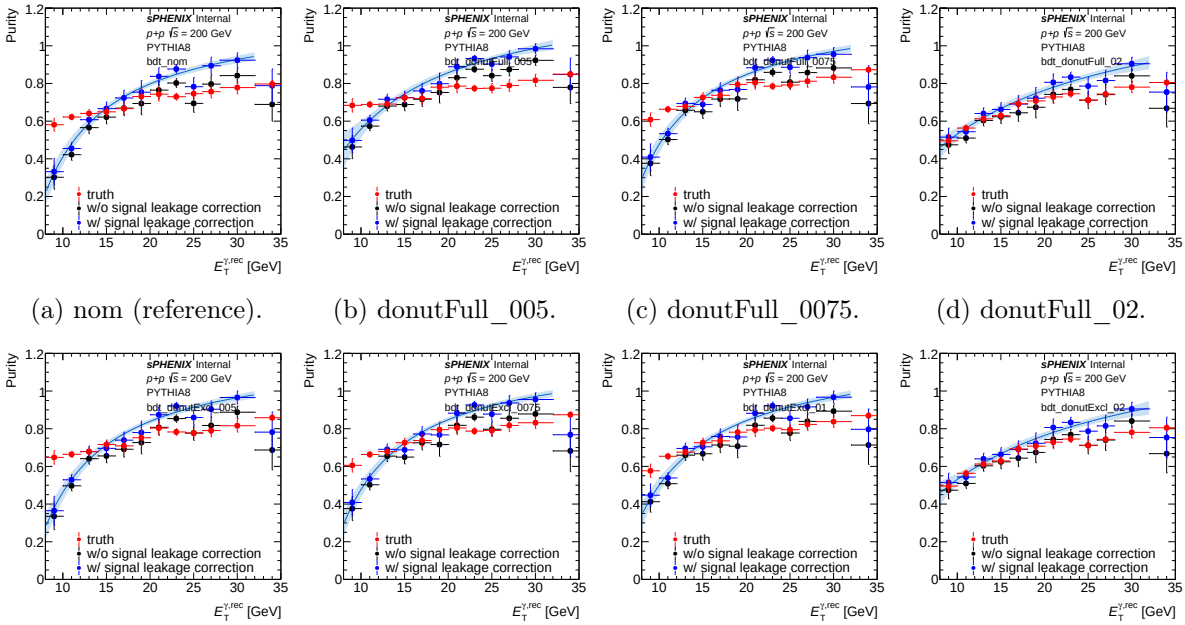
- **Outer cone includes HCal.** The donut is not pure EMCal. A pure-EMCal donut requires new inner-ring branches `cluster_iso_{005,0075,01,02}_emcal` at matching threshold (logged in `reports/anatreemaker_reprocess_TODO.md`).
- `donutFull_01` **skipped** pending upstream `cluster_iso_01` branch (all-calo, 120 MeV, $R < 0.1$).
- **Degraded mirrors** (`Cluster_rbr.C`, `EtaMigrationStudy.C`, `compare_energy_response.C`) silently use the full outer cone without inner subtraction for cases 4/5 since they do not read `iso_emcalinnerr`. Do not run donut variants through these diagnostic utilities expecting donut semantics.
- **MC iso scale/shift** set to 1.0/0.0 for all donut variants because the nominal 1.2/0.1 was tuned against full-cone topo04. Retuning would be needed if these ever became systematic variants.
- **The low-p_T closure deviation is present in nominal too.** This is a pre-existing effect of the ABCD method at the binning edge, independent of the iso definition. Worth understanding before any purity systematic is claimed.

Reviewer pass matrix

Artifact	Physics	Cosmetics	Re-derivation	Fix-valida
`use_topo_iso=4,5` switch	PASS (2 WARN util)	N/A	N/A	PASS
donut truth-purity scan PDFs	N/A	PASS (2 WARN)	PASS	N/A
donut closure PDFs	N/A	(not yet reviewed)	PASS	N/A
80 % iso cuts	N/A	N/A	PASS	PASS
`g_purity_truth` (7 variants)	N/A	N/A	PASS	PASS
`g_mc_purity_fit_ratio` (7 variants)	N/A	N/A	PASS	PASS

Per-variant MC purity closure

Each panel below shows, for one variant, the ABCD-derived purity on MC (points with error bars), the fit to the ABCD points (solid line), the signal-leakage-corrected ABCD purity, and the truth-matched purity (red markers). For a closure-good variant, the ABCD points/fit should track the truth markers; deviations mark where the ABCD subtraction biases the purity estimate at that p_T. Nominal topo04 iso (`bdt_nom`) is included as a reference.



(a) nom (reference). (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.

(e) donutExcl_005. (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 3: Per-variant MC purity closure plots. Each panel: ABCD-derived purity points (from `gpurity`/`gpurity_leak`), the ABCD fit, and truth-matched purity (`g_purity_truth`). The quality of agreement between ABCD and truth across p_T is the closure check summarized in the previous section's ratio table. `donutFull_02` and `donutExcl_02` show the tightest truth–ABCD agreement, especially at low p_T.

Final cross section per variant

Donut / nominal ratio

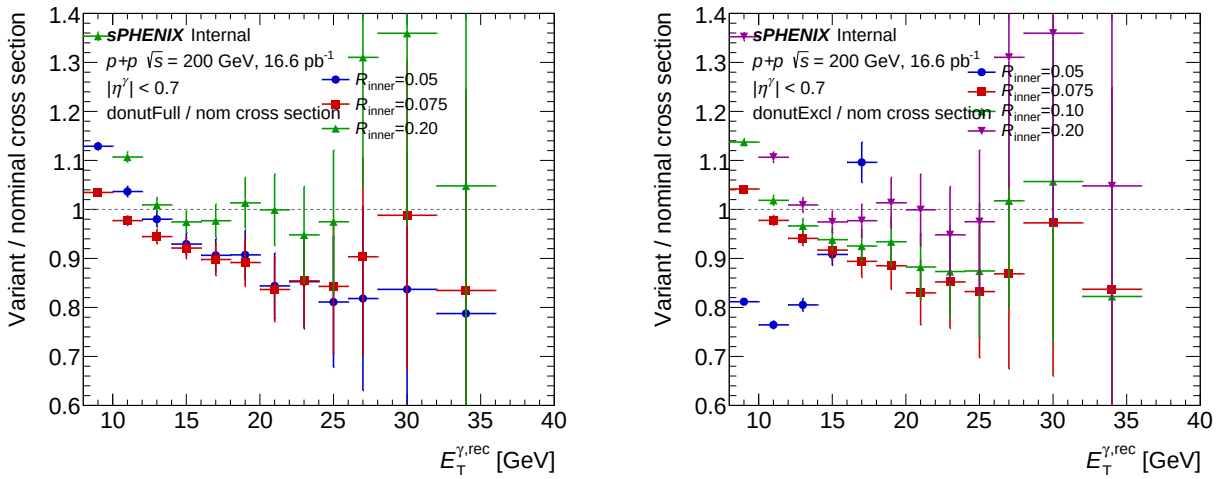
The following ratio is defined per reco p_T bin on the unfolded signal yield `h_unfold_sub_result` (which is the background-subtracted, unfolded yield used to compute $d^2\sigma/d\eta\, dp_T$): $r = \text{yield}_{\text{donut}}/\text{yield}_{\text{nom}}$.

Table 4: Donut / nominal unfolded-yield ratio at selected p_T bins. Mean absolute deviation from unity averaged over the 12 reco p_T bins.

Variant	$r(9)$	$r(15)$	$r(21)$	$r(28)$	$r(34)$	$\langle\lvert 1-r\rvert\rangle$
donutFull_005	1.129	0.929	0.844	0.818	0.788	0.130
donutFull_0075	1.035	0.921	0.837	0.903	0.835	0.097
donutFull_02	1.354	0.974	0.999	1.310	1.048	0.138
donutExcl_005 (anomalous)	0.812	0.908	1.937	**−6.02**	**−0.04**	1.665
donutExcl_0075	1.042	0.917	0.830	0.869	0.837	0.104
donutExcl_01	1.137	0.938	0.882	1.018	0.822	0.096
donutExcl_02	1.352	0.974	0.999	1.310	1.048	0.137

donutExcl_005 anomaly. This variant yields negative unfolded signal at $p_T \geq 28\,\text{GeV}$, implying the ABCD background estimate exceeds the tight+iso count in those bins. Looking at the raw `h_unfold_sub_result` across p_T: values at $p_T = 22$–26 are anomalously large (2–6× nominal) before the bin-11 sign flip to −5.66 and bin-12 −0.81. Truth purity for the same variant is healthy (0.65–0.86), so the pathology is in yield extraction, not purity. Likely an interaction between the tight-iso count and the R-factor estimator at low statistics with the tightest inner R + ownership-excluded outer cone. Recommend: re-run with higher MC statistics (more condor jobs

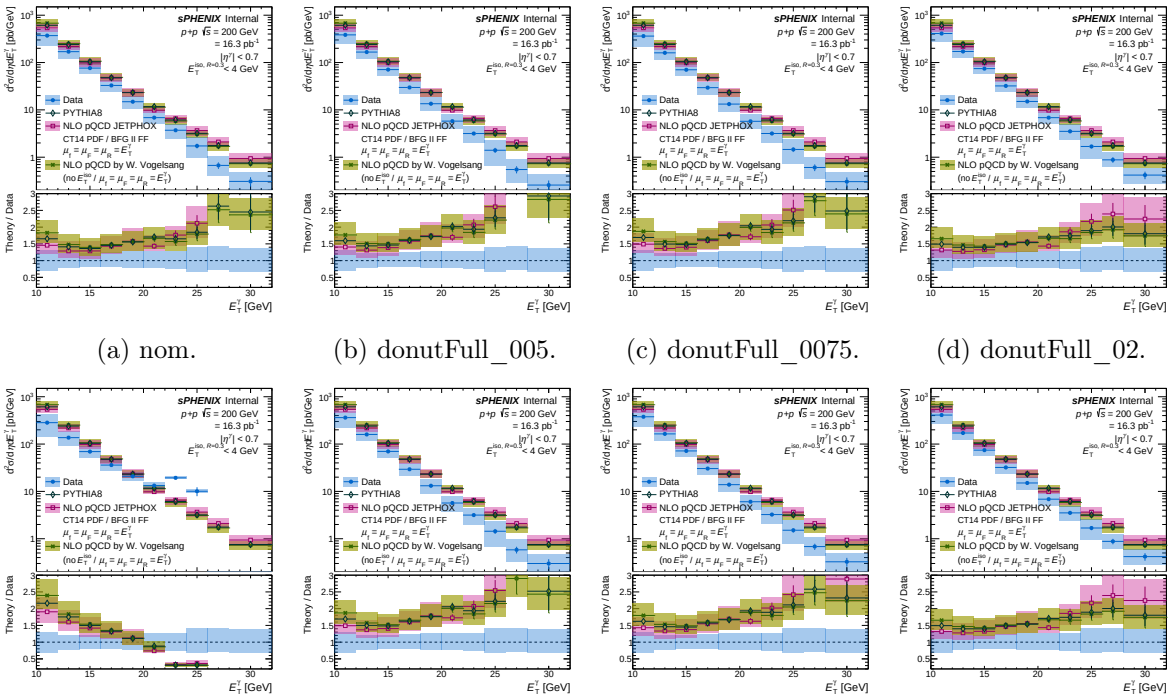
per sample) or drop this variant from the scan.



(a) donutFull / nominal. (b) donutExcl / nominal.

Figure 4: Unfolded-yield ratio (donut / nominal) vs. reco p_T. Dashed line at unity. Excluding `donutExcl_005`, variants agree with nominal to $\sim$ 10–14% on $\langle|1-r|\rangle$. The `R_inner`=0.20 variants diverge upward at low p_T (ratio up to 1.35) and stay within $\pm 5\%$ of nominal at mid-p_T. `donutExcl_005` is plotted but pT $\geq$ 26 GeV points fall off-scale (negative / $\gg$ 1.4).

Per-variant final cross section



(a) nom. (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.

(e) donutExcl_005 (anomalous). (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 5: Per-variant final isolated-photon cross section $d^2\sigma/d\eta\, dp_T$ vs. p_T^γ, produced by `plot_final_selection.C`. Nominal topo04 iso (first panel) for reference. The `donutExcl_005` panel shows the high-p_T anomaly noted above.